

THOMAS STECKMANN

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EDUCATION

University of Maryland, College Park: Physics PhD Student
Lanczos Fellow, Joint Center for Quantum Information and Computer Science College Park, MD
Aug. 2022 – 2027 (Expected)

North Carolina State University: 4.0 GPA - Physics B.S. | Mathematics B.S.
University Honors; Math Honors; Sigma Pi Sigma; Phi Beta Kappa; Park Scholar Raleigh, NC
Aug. 2018 – May 2022

RESEARCH

Error Mitigation Methods for Quantum Algorithms (PhD research) Ongoing
University of Maryland, College Park MD College Park, MD

- Exploring the integration of quantum algorithms and error mitigation strategies for optimizing the execution of quantum algorithms on noisy hardware.
- Working knowledge of error mitigation strategies and techniques for Hamiltonian Simulation in condensed matter systems

Hamiltonian Fast-Forwarding Algorithms (DOE SULI program) Summer 2020 - Spring 2021
Oak Ridge National Laboratory and North Carolina State University Oak Ridge, TN

- Extended methods in unitary matrix synthesis for quantum computers to allow for fast-forwarding long time scale dynamics for Hamiltonian simulation. Work with the Cartan decomposition of dynamical Lie groups generated by a Hamiltonian.
- Published in Physical Review Letters [Fixed Depth Hamiltonian Simulation via Cartan Decomposition](#)
- Developed and published a python package to simplify the implementation of the decomposition algorithm and to encourage exploration into applications beyond the expertise of the authors: [Cartan Quantum Synthesizer](#)

Algorithms for Error-Prone Quantum Computers (SULI/NCSU Honors) Summer 2021 - Present
Oak Ridge National Laboratory and North Carolina State University Oak Ridge, TN / Raleigh, NC

- [Lead author work](#) demonstrating the application of Cartan-based fast-forwarding circuits for use in dynamical mean-field theory to map the Mott-insulator phase transition in the thermodynamic limit of the Hubbard Model
- Developed or implemented a variety of error mitigation methods—result post-processing, randomized Cartan compiling, and measurement error mitigation—exploiting the structures of Cartan decomposition and the conserved quantities in the simulation of the Anderson Impurity Model
- Successfully recovered physical dynamics in the presence of significant quantum noise via use of noise robust algorithms.

Electron Transport in Organic Transistors June 2018 – June 2020
North Carolina State University – Organic and Carbon Electronics Lab Raleigh, NC

- Developed methodology to investigate the micro-structural influences on electron transport mobility in organic, field effect transistors
- Demonstrated and explained an efficient fabrication method using floated polymer films as a means to preserve high quality charge transport in films down to only two molecular layers. The method opens up possibilities for fabrication of material efficient, flexible, and transparent transistors
- [Published in Advanced Electronic Materials](#). Results presented at the Material Research Society Fall 2020 meeting

Optimization Methods for Cartan Decomposition Spring/Fall 2021
North Carolina State University Raleigh, NC

- Exploring quantum variational cost functions for optimization in Cartan Decomposition
- Solved the parameter minimization problem in Cartan decomposition using parameter flows in Lax dynamics, allowing for finding a solution through numerical integration of an ordinary differential equation to reduce computational complexity – led to publication
[Lax dynamics for Cartan decomposition with applications to Hamiltonian simulation](#)

PUBLICATIONS AND PRESENTATIONS

Publications:

- **Thomas Steckman**, De Luo, Yu-Xin Wang, Sean R Muleady, Alireza Seif, Christopher Monroe, Michael J Gullans, Alexey V Gorshkov, Or Katz, and Alexander Schuckert. "Error mitigation of shot-to-shot fluctuations in analog quantum simulators," (July 2025) [arXiv:2506.16509](https://arxiv.org/abs/2506.16509)
- Norman Hogan*, Efehan Kökcü*, **Thomas Steckmann**, Liam P Doak, Carlos Mejuto-Zaera, Daan Camps, Roel Van Beeumen, Wibe A de Jong, and AF Kemper. "A quantum computing approach to efficiently simulating correlated materials using impurity models and dynamical mean field theory," (August 2025) [arXiv:2508.05738](https://arxiv.org/abs/2508.05738)
- **Thomas Steckmann**, Trevor Keen, Efehan Kökcü, Alexander F. Kemper, Eugene F. Dumitrescu, and Yan Wang. "Mapping the metal-insulator phase diagram by algebraically fast-forwarding dynamics on a cloud quantum computer," (June 2023) [Physical Review Research \(Open Access\)](https://doi.org/10.1103/PhysRevResearch.5.023111)
- **Thomas Steckmann**, Indunil Angunawela, Somayeh Kashani, Youqin Zhu, Masrur M. Nahid, Harald Ade, and Abay Gadisa. "Ultrathin P(NDI2OD-T2) Films with High Electron Mobility in Both Bottom-Gate and Top-Gate Transistors." (March 2022) [Advanced Electronic Materials](https://doi.org/10.1002/aelm.202100000).
- Efehan Kökcü, **Thomas Steckmann**, JK Freericks, Eugene F. Dumitrescu, and Alexander F. Kemper, "Fixed Depth Hamiltonian Simulation via Cartan Decomposition," (August 2022) [Physical Review Letters \(available on arXiv\)](https://arxiv.org/abs/2208.00000).

Software:

- **Thomas Steckmann**, Efehan Kökcü. (Summer 2021) [Cartan Quantum Synthesizer](https://github.com/steckmann/CartanQuantumSynthesizer)

Presentations:

- "<https://www.youtube.com/watch?v=QbLDda4t7Dg>" **Contributed Talk**, QSim 2025 | Summer 2025
- "Error Mitigation for Analog Simulators" *American Physical Society, March Meeting* | Spring 2025
- "Error Mitigation for Analog Simulators" **Invited Talk**, *Institute for Robust Quantum Simulation, Annual Meeting* | Summer 2024
- "Error Mitigation for Analog Simulators" *American Physical Society, March Meeting* | Spring 2024
- "[Simulating the Mott transition on a noisy digital quantum computer via Cartan-based fast-forwarding circuits](https://arxiv.org/abs/2203.00000)" *American Physical Society, March Meeting* | Spring 2022
- "[Scalable Fabrication of High Mobility Monolayer OFETs Using Floating Film Transfer](https://arxiv.org/abs/2010.00000)" Poster, *Material Research Society* | Fall 2020
- "Simulating Quantum Systems on Dubious Quantum Computers" **Invited Award Talk**, *NC State McCormick Symposium* | Spring 2022
- "Constant Depth Exact Time Evolution of Spin Systems Based on Cartan Decomposition," *National Council For Undergraduate Research* | 2021
- 2nd place, "Efficient Quantum Simulation on a Quantum Computer," *NC State McCormick Symposium* | 2021
- 3rd place: "Aggregation Mechanisms in Low Temperature Ultrathin Water-Floated Films" Research Proposal: [Future of Materials IV Workshop](https://arxiv.org/abs/2003.00000) | 2020
- "A Highly optimized quantum circuit for simulating two-site dynamical mean-field theory on noisy quantum hardware" Joint Quantum Computation and Quantum Information Technical Talks, *Oak Ridge National Laboratory* | 2021
- Honorable Mention, "High Efficiency Semiconducting Polymer Self-Assembly in Organic Transistors" *NC State McCormick Symposium* | 2020
- "Lax Dynamics for Quantum Computing: Applications to Cartan Decomposition" *Mathematics Honors Presentation*, *NC State* | April 2022

TECHNICAL SKILLS AND RELEVANT COURSEWORK

Programming: Python, Java, Mathematica, HTML, CSS, $\LaTeX$, Git

Libraries/Packages: Qiskit, NumPy, Scipy, Matplotlib, Numba, OpenFermion

Featured Course Work: Quantum Error Correction |— Quantum Hardware and Metrology | Classical Mechanics | Quantum Mechanics | Electrodynamics [Mathematical Foundations of Quantum Computation](https://arxiv.org/abs/2003.00000) | Linear Algebra | Complex Analysis | Computational Physics | Modern Algebra | [Cryptography](https://arxiv.org/abs/2003.00000) | Java Software Development

QUANTUM COMPUTING

Undergraduate School on Experimental Quantum Information Processing | *University of Waterloo* Summer 2021

- Virtual summer program featuring lectures on the theory and implementation of quantum algorithms, qubit architectures, and quantum communication protocols

Quantum Ideas Summer School | *Duke University STAQ collaboration* Summer 2019

- Introduction to quantum computing hardware approaches including superconducting and trapped ion systems
- Exploration of variational algorithms and error correcting codes

Quantum Computing Hackathons with Qiskit 2020

- Qiskit Hackathon Global: With a team of five students, developed an easy to use function to evaluate the Quantum Volume of IBM hardware during a 24 hour invite only event
- [NC IBM Q Summer Jam](#): Implemented an algorithm for quantum PageRank using Quantum Random Walk

COMMUNITY ENGAGEMENT AND LEADERSHIP DEVELOPMENT

Quantum Information Club at NC State | *Co-founder, President* March 2020 – 2022

- Approachable programming in quantum computing for undergraduate students with a range of math, science, and engineering backgrounds by working closely with community partners such as the NC State Q Hub and IBM Qiskit, and university groups at UNC Chapel-Hill, Duke, Georgetown, and Georgia Tech
- Develop and teach interactive lessons and coding tutorials on the fundamentals of quantum algorithms and quantum information

Park Scholarships 2018 - 2022

- Four year, merit based scholarship awarded for excellence in Leadership, Service, Character, and Scholarship
- Development courses focusing on leadership, civic engagement, and diversity

CrowdSolv web development 2019 - 2020

- Volunteer front end development (HTML, CSS, Bootstrap, Javascript) for CrowdSolv, a non-profit building a collaborative platform for crowd-sourced problem solving

Habitat for Humanity | Works Project Officer 2019 - 2022

- Regular volunteer with Habitat Wake and the NCSU chapter to promote affordable housing access and work in home construction
- As a club officer, I assist with fundraising efforts, facilitate student involvement, and work with construction leaders to guide and train new volunteers

ACADEMIC SERVICE

Conference and Journal Reviews

- Theory of Quantum Computing; Young Quantum Information Scientists; NPJ Computational Materials; Physics Review Research

Seminar Organizer 2023-2025

- Logistics and scheduling for weekly departmental quantum seminar

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- Develop and teach interactive lessons and coding tutorials on the fundamentals of quantum algorithms and quantum information

AWARDS AND HONORS

QulCS Lanczos Graduate Fellowship 2022 - 2024

- Two-year fellowship with RA support within the Joint Center for Quantum Information and Computer Science

Park Scholarship 2018 - 2022

- Undergraduate, four-year, full cost-of-living merit-based scholarship. Includes four year program on leadership and civic development
- **Outstanding Research Award** (2022) - College of Sciences, North Carolina State University
- **Rodney I. McCormick Award for outstanding research** (2022) - Department of Physics, North Carolina State University